Università di Pisa
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Master's Degree in Physics

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Chapter 1

Introduction

Chapter 2

The dataset

2.1 Source

The flights come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition of the French free-flight federation (FFVL). Pilots declare their flights online, and each entry carries the flight's metadata and, in most cases, its GPS track. Paragliders and hang gliders have separate archives; a larger share of hang-glider entries has no track.

A season runs from 1 September to 31 August. We use the paraglider seasons from 1999–2000 and the hang-glider seasons from 2001–2002, up to 2025–2026. For each season we download the list of declared flights, kept verbatim as provenance, and the tracks it links. A saved list is never replaced, so each one is a snapshot of the day it was downloaded: the 2025–2026 lists were saved around the end of June 2026 and miss the rest of that season.

2.2 Tracks and metadata

Tracks are IGC files [1], plain-text records of the flight. Each `B` record is one *fix*: UTC time, latitude and longitude, a validity flag, and two altitudes, the pressure altitude from the barometer and the GNSS altitude from the satellite fix. The sampling interval depends on the recorder and need not be constant within a flight. Some recorders log no pressure altitude.

The metadata give the date, the pilot, the take-off and landing sites, the wing model and class, the scored distance and the scoring category of the declared route (free distance, out-and-return, flat or FAI triangle and a few others). The category describes the declared route only. A few paraglider entries have an incomplete date, such as `0000-00-00`, which excludes them from any comparison by date.

2.3 Glider categories

Wing design changes speed, glide performance and turning geometry, so paragliders and hang gliders are analysed separately [2]. A paraglider is a flexible canopy; a hang glider has a framed wing, flexible or rigid.

The wing class follows a different system in each discipline. Paraglider classes are those of the EN 926-2 standard, A to D, written with their equivalent in the older LTF scale ("C ou 2" reads "C or 2"), plus the CIVL Competition Class and a few other labels, such as tandem or uncertified wings. EN classes rank passive safety and the skill needed to recover from a collapse, and say nothing direct about speed or glide ratio [3]. Hang-glider classes are those of

Table 2.1. Paraglider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.2. Hang-glider flights per season.

Season	Flights	With track	Downloaded	Downloaded (% of tracks)
2001–2002	248	0	0	–
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

the FAI [4]: Class 1, with its Sport sub-class, for wings controlled by weight shift alone, and Classes 2 and 5 for rigid wings with movable control surfaces.

Chapter 3

Trajectory pre-processing

The pre-processing turns each IGC file into a trajectory fit for analysis. This chapter describes its steps in the order they run, one section per step. Some steps delete single fixes; others judge the whole flight and keep or drop it.

3.1 Reading the fixes

Each `B` record gives a fix $(t_i, \varphi_i, \lambda_i)$, with latitude and longitude in degrees, together with the validity flag and the two altitudes h_i^{baro} and h_i^{GNSS}. A record that does not follow the fixed layout of the format, or whose time of day or position is impossible, is discarded. A missing or unreadable altitude, including the zero the format uses for a missing reading [1], is marked as missing, and the fix keeps its time and position.

Time is measured in seconds from the first fix. Records carry only the time of day, so a flight crossing UTC midnight would jump back by a day; a day is added when the clock passes from the last hour of one day to the first hour of the next. Other backward steps and repeated times are left to the clock rule of Section 3.3. Well-formed but implausible fixes, such as position spikes, are kept as well: the parser discards only what cannot be read. The validity flag, which marks fixes without a three-dimensional GNSS solution, is kept but not used at this stage.

3.2 Choice of altitude channel

Some recorders log no pressure altitude, so we use GNSS altitude for every flight, $z = h^{\text{GNSS}}$. Its vertical datum is not uniform across recorders: the IGC specification prescribes height above the WGS84 ellipsoid [1], the CIVL specification for hang gliders and paragliders height above the geoid [5]. A constant offset cancels in altitude increments, but comparing absolute altitudes across flights would need a datum audit.

Let N be the number of fixes read from the file, before any is deleted, and P the set of those with a GNSS altitude. The flight is kept only if

$$\frac{|P|}{N} \geq 0.95 \qquad \text{and} \qquad \max_{i \in P} h_i^{\text{GNSS}} - \min_{i \in P} h_i^{\text{GNSS}} \geq 30\,\text{m}. \tag{3.1}$$

The first condition rejects incomplete channels; the second rejects channels stuck at one value, and flights with almost no vertical excursion. Fixes flagged as lacking a three-dimensional solution count as present if they carry an altitude. Pressure altitude is never substituted for a missing GNSS value.

The barometer is checked with the same two conditions on h^{baro}, with thresholds 0.95 and 30 m. A flight whose barometer passes is marked, so that later cleaning steps can use pressure altitude as independent evidence; the mark does not decide whether the flight is kept.

3.3 Clock

A recorder can write a time equal to or earlier than the one before it. On the fixes as read, the clock rule keeps the first fix, and fix i only if its time is later than that of every fix before it,

$$t_i > \max_{j<i} t_j. \tag{3.2}$$

Of several fixes with the same second, the first is kept. A backward step is deleted, and so is every fix after it until the clock passes the latest time already kept. The times left are strictly increasing, so every step between consecutive fixes has a positive duration (Section 3.5).

3.4 Repeated positions

Some recorders repeat their last position while the clock runs, so a fix can carry the coordinates of the one before it without being a new measurement. On the fixes left by the clock rule, numbered again in order, fix i is deleted if

$$\varphi_i = \varphi_{i-1} \qquad \text{and} \qquad \lambda_i = \lambda_{i-1}. \tag{3.3}$$

The comparison is exact: the two records carry the same digits. Altitudes are not compared. In a run of fixes at the same position only the first is kept.

3.5 Fast tracks

A free flight spends part of its time climbing in thermals at a low ground speed, so the typical speed of its track stays well below the top speed of the wing. A track with a higher typical speed comes from something else, such as a powered aircraft or a file with compressed times, and the whole flight is dropped.

Let n be the number of fixes left by the two previous rules. Between fix i and fix $i+1$ the horizontal distance is taken on a sphere of radius $R = 6{,}371$ km, with the haversine formula

$$d_i = 2R \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos\varphi_i \cos\varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \tag{3.4}$$

with the angles in radians, and the step speed is

$$v_i = \frac{d_i}{t_{i+1} - t_i}, \qquad i = 1, \ldots, n-1, \tag{3.5}$$

. The clock rule makes every denominator positive. The typical speed of the flight is the median of its step speeds,

$$\tilde{v} = \mathrm{median}\{v_1, \ldots, v_{n-1}\}, \tag{3.6}$$

and the flight is kept if $\tilde{v}$ does not exceed 70 km/h for paragliders and 110 km/h for hang gliders. A flight with fewer than two fixes has no step and is kept. The rule deletes no fix, and the median is not moved by a few wrong steps, such as a position spike: those are left to a later step.

The thresholds were read off the distribution of $\tilde{v}$ over all flights (Figure 3.1). In both disciplines the bulk of the distribution ends below the threshold, and beyond it only a sparse tail of single flights remains. Split by wing class (Figures 3.2 and 3.3), the bulk of every class stays below the threshold. The faster shoulder of the hang-glider distribution comes from the rigid Class 2 wings.

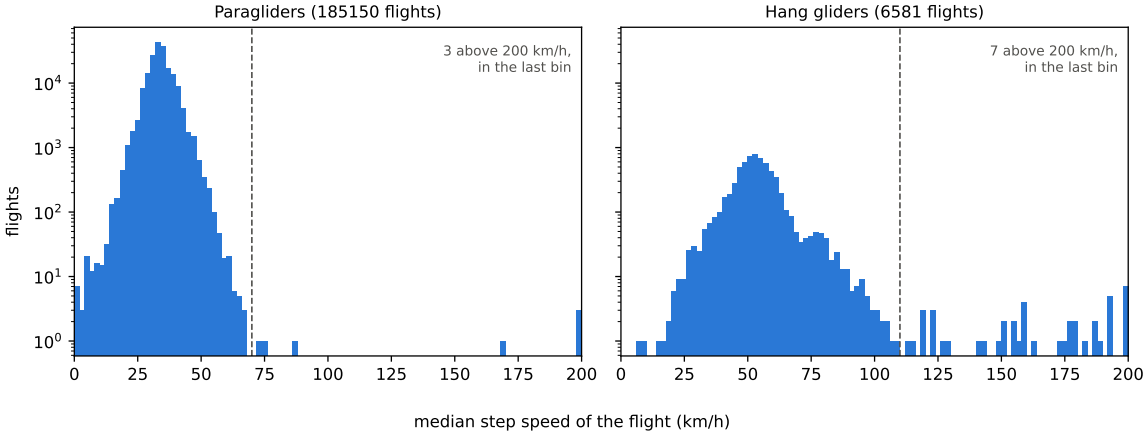


Figure 3.1. Median step speed $\tilde{v}$ (Equation (3.6)) of every flight past the altitude channel, computed after the clock and repeated-position rules. Bins of 2 km/h, logarithmic vertical axis; flights above 200 km/h fall in the last bin. Dashed line: threshold, 70 km/h for paragliders and 110 km/h for hang gliders.

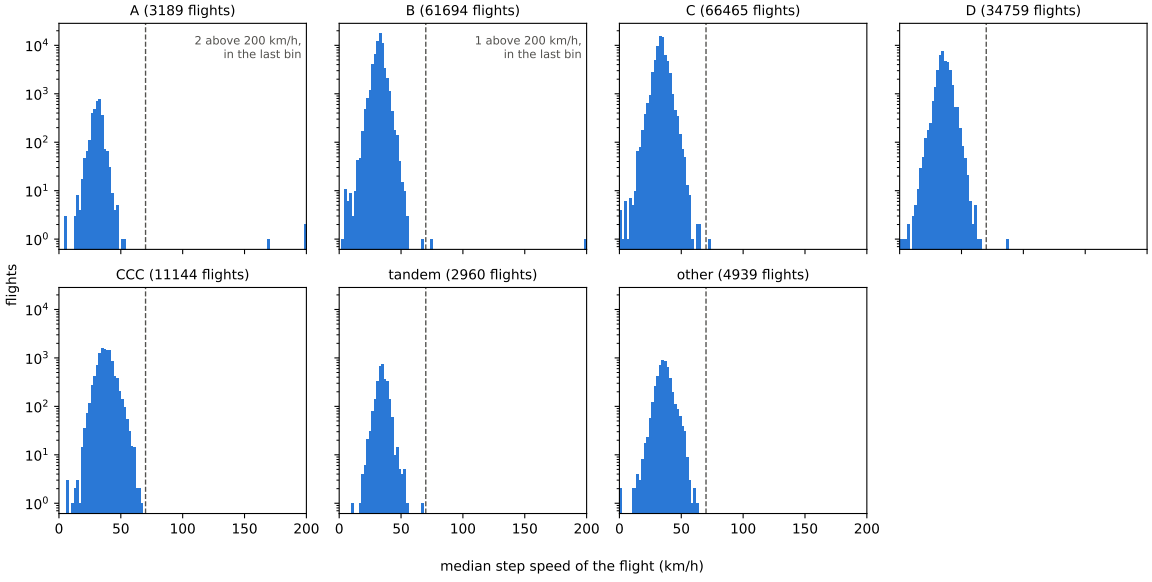


Figure 3.2. As Figure 3.1, for paragliders by wing class (Section 2.3): EN classes A to D, the CIVL Competition Class (CCC), tandem wings, certified or not, and every other label.

3.6 Horizontal excursions

A step whose speed v_i, Equation (3.5), exceeds what any glider of the discipline can fly is a recorder error, but the step alone does not say what went wrong. Either the track has left its path and comes back to it, as in a spike, or it goes on from the new position, as in a jump. The excursion rule tells the two apart by looking ahead: it deletes the fixes of an excursion

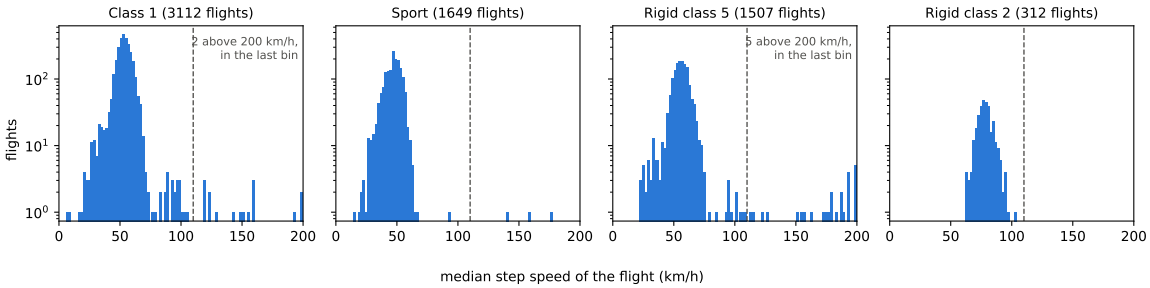


Figure 3.3. As Figure 3.1, for hang gliders by wing class (Section 2.3): Class 1, its Sport sub-class, and the rigid Classes 5 and 2. Flights of any other class are not shown.

that comes back within a time W, and splits the flight where the track does not come back. It has two parameters: the top speed $v_{\max}$, read off the data, and W, set by hand and tested at the end of the section.

The rule scans the fixes left by the previous steps in order, against an anchor a, the last fix accepted. Fix j is reachable from the anchor if

$$d(a, j) \leq v_{\max}\,(t_j - t_a), \tag{3.7}$$

with $d(a, j)$ the distance of Equation (3.4) between the two fixes. A reachable fix is accepted and becomes the anchor. An unreachable fix j opens a block, and the rule looks for a later fix k reachable from the anchor within $W = 20\,\text{s}$ of the opening, $t_k - t_j \leq W$. If there is one, the track went out and came back: fixes j to $k-1$ are deleted and k becomes the anchor. If there is none, the track does not come back: the flight is split between the anchor and fix j, and a new segment starts from j. A block still open at the end of the track is deleted. The rule thus leaves the fixes it accepted and the places where the flight is split, which a later step uses as the boundaries of its segments. Figure 3.4 shows both cases.

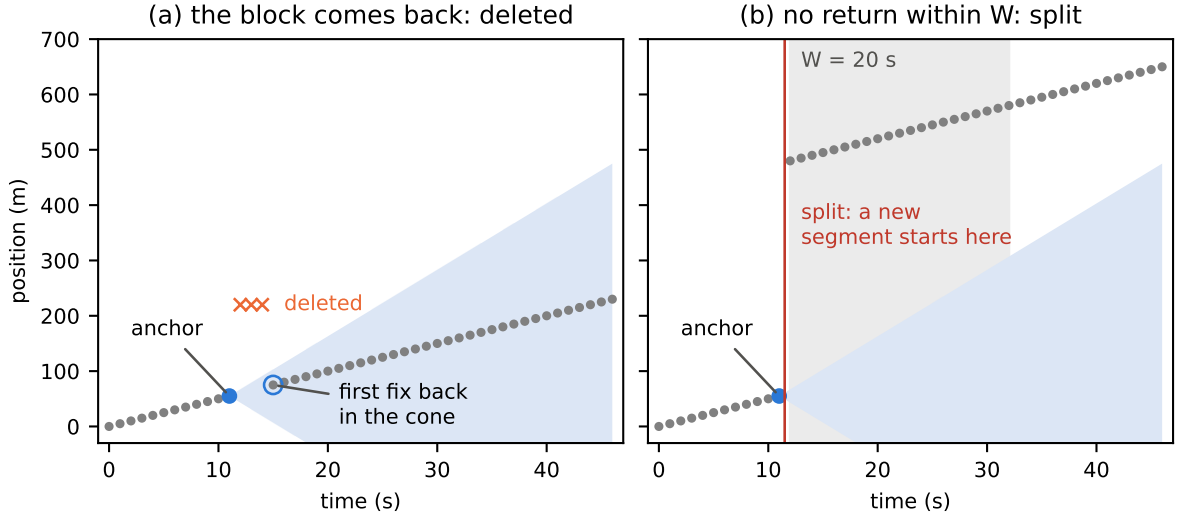


Figure 3.4. The excursion rule on two synthetic tracks along one coordinate, one fix per second, with a top speed lowered for legibility. Grey dots: fixes kept. Blue dot: the anchor, the last fix accepted before the block. Shaded area: the positions x reachable from the anchor, $|x - x_a| \leq v_{\max}(t - t_a)$, Equation (3.7) along one coordinate. (a) Three fixes leave the cone and the next one comes back within W: the three are deleted (crosses). (b) The track jumps and stays there: no fix comes back within $W = 20\,\text{s}$ of the opening of the block (grey band), so the flight is split (vertical line) and a new segment starts from the block.

The first anchor of each segment, at the start of the track or after a split, is the first fix whose step to the next fix satisfies Equation (3.7); the fixes before it are deleted. An anchor off the track would make the good fixes after it unreachable, while an isolated wrong fix disagrees with both its neighbours and so is never chosen. The price is that a good fix followed by a spike is deleted with it.

The top speed must lie above the speed of real flight, and the rule tells which speed of real flight matters. Suppose a stretch of real flight is faster than $v_{\max}$: the fix that ends its first fast step opens a block, after an anchor a that is itself on the track. Over a window of length W, the speed from fix i is

$$u_i = \frac{d(i,\ell)}{t_\ell - t_i}, \qquad \ell = \min\{k : \ t_k - t_i \geq W\}, \tag{3.8}$$

with d as in Equation (3.7). For $i = a$ the window covers real flight, and unless it spans a gap, its last fix ℓ comes within W of the opening of the block. That fix is reachable from the anchor if $u_a \leq v_{\max}$. A top speed above every window speed of real flight therefore never splits real flight: a stretch of real flight faster than $v_{\max}$ can lose its fixes, but the rule finds a reachable fix in time (Figure 3.5).

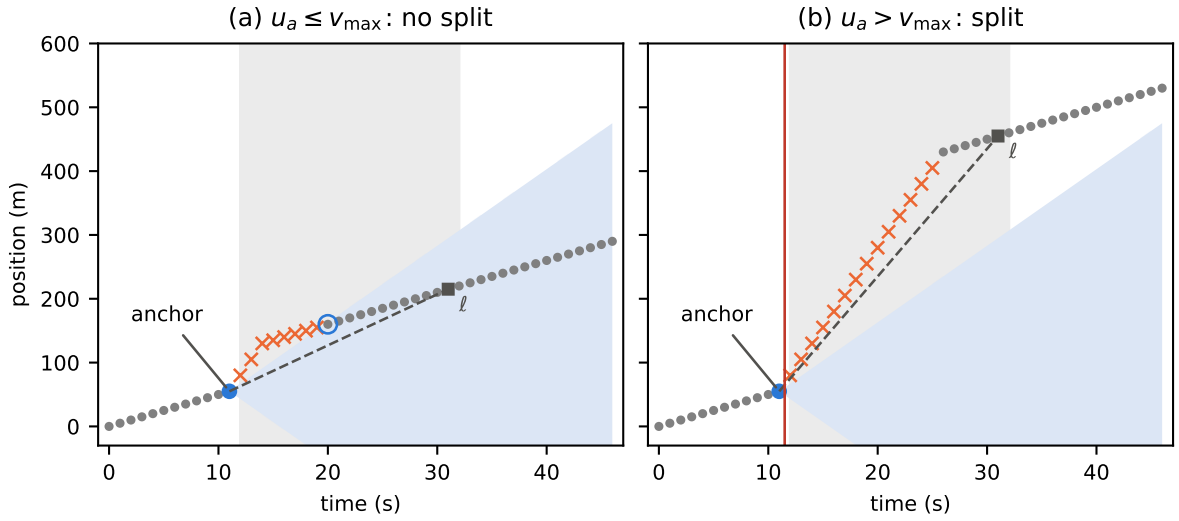


Figure 3.5. Real flight faster than the top speed, on synthetic tracks drawn as in Figure 3.4. A burst of fast steps opens a block after the anchor a; grey band: the time W the rule waits for a reachable fix. Dashed line: from the anchor to fix ℓ, the end of the window of length W from the anchor, Equation (3.8); its slope is the window speed u_a. (a) A short burst: $u_a \leq v_{\max}$, so ℓ lies in the cone and within the band. The rule finds a reachable fix (circle), ℓ at the latest, and deletes the fixes before it (crosses), but does not split the flight. (b) A burst that lasts most of the window: $u_a > v_{\max}$, no fix is reachable within W, and the rule splits real flight (vertical line). The fast fixes are deleted too, because the new segment starts at the first fix whose step to the next is reachable.

Computed from every fix, the window speed also follows real flight much further than the step speed. A single fix displaced by a distance D changes u_i by at most D/W, against $D/\Delta t$ for the two steps that touch it. Both distributions fall steeply (Figure 3.6, lower row); the step speed then turns into a long tail of errors, the window speed only later and much lower.

The window speed still carries errors that last longer than W, such as a jump that never comes back. Split by wing class (Figure 3.6, upper row), real flight depends on the wing: each class falls off in the order of its performance. The errors do not depend on the wing, and beyond the fall-off the curves of all classes merge into one common tail. Real flight therefore ends where the fastest class reaches that tail. We set $v_{\max}$ there: 100 km/h for paragliders and 175 km/h for hang gliders.

To see how much the output depends on W, the rule was run on every flight kept by the previous steps with W halved and doubled, 10 s and 40 s, and $v_{\max}$ unchanged (Table 3.1). The share of fixes deleted barely changes, so almost every excursion that comes back does so within the shortest window. The rule splits the flight at every excursion that does not come back within W, so a shorter W splits more flights: the share of flights split roughly doubles when W is halved and halves when it is doubled, and stays a small fraction of the flights for every window. Any W also treats as a return a jump shorter than about $v_{\max}W$ after which the track stays where it landed, and deletes the fixes in between, so a longer W bridges larger jumps. Across this range the choice affects few flights, and we keep the middle value, which balances splits against bridged jumps.

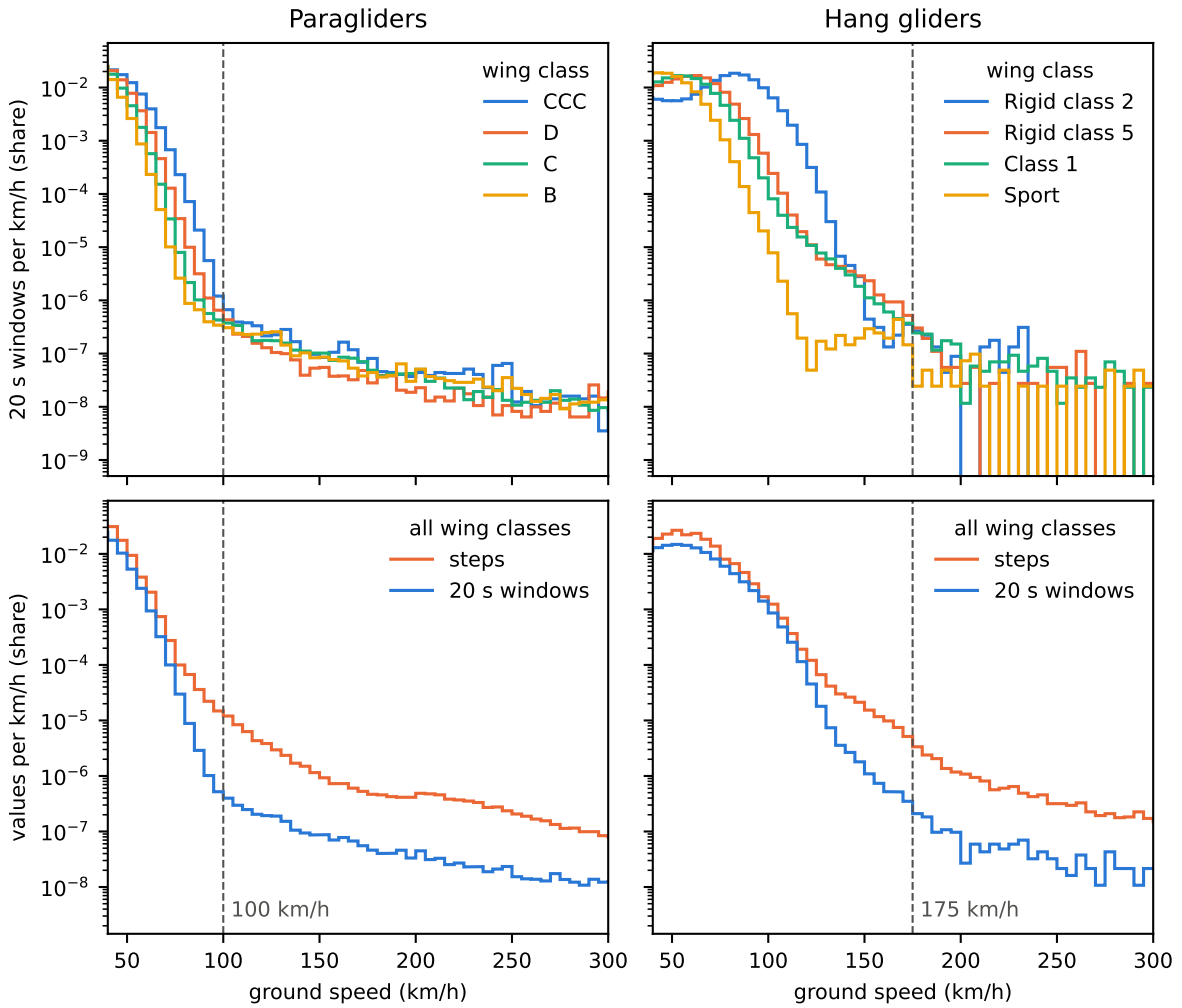


Figure 3.6. Ground speed in the flights kept by the previous steps. Upper row: speed over windows of length $W = 20$ s, Equation (3.8), for four wing classes of each discipline (Section 2.3). Lower row, all classes together: speed of single steps, Equation (3.5), and over windows. Each curve is the share of its values per km/h, in bins of 5 km/h, on a logarithmic axis; values beyond the axis count in the share. Dashed line: the top speed $v_{\max}$, 100 km/h for paragliders and 175 km/h for hang gliders.

Table 3.1. The excursion rule with the window $W = 20\,\mathrm{s}$ and with half and twice W, in the flights kept by the previous steps. Flights split: share of the flights split at least once. Fixes deleted: share of the fixes the rule receives that it deletes.

	Paragliders (185,321 flights)		Hang gliders (6,536 flights)	
W (s)	Flights split (%)	Fixes deleted (%)	Flights split (%)	Fixes deleted (%)
10	2.08	0.032	1.27	0.014
20	0.96	0.033	0.72	0.015
40	0.45	0.034	0.32	0.016

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